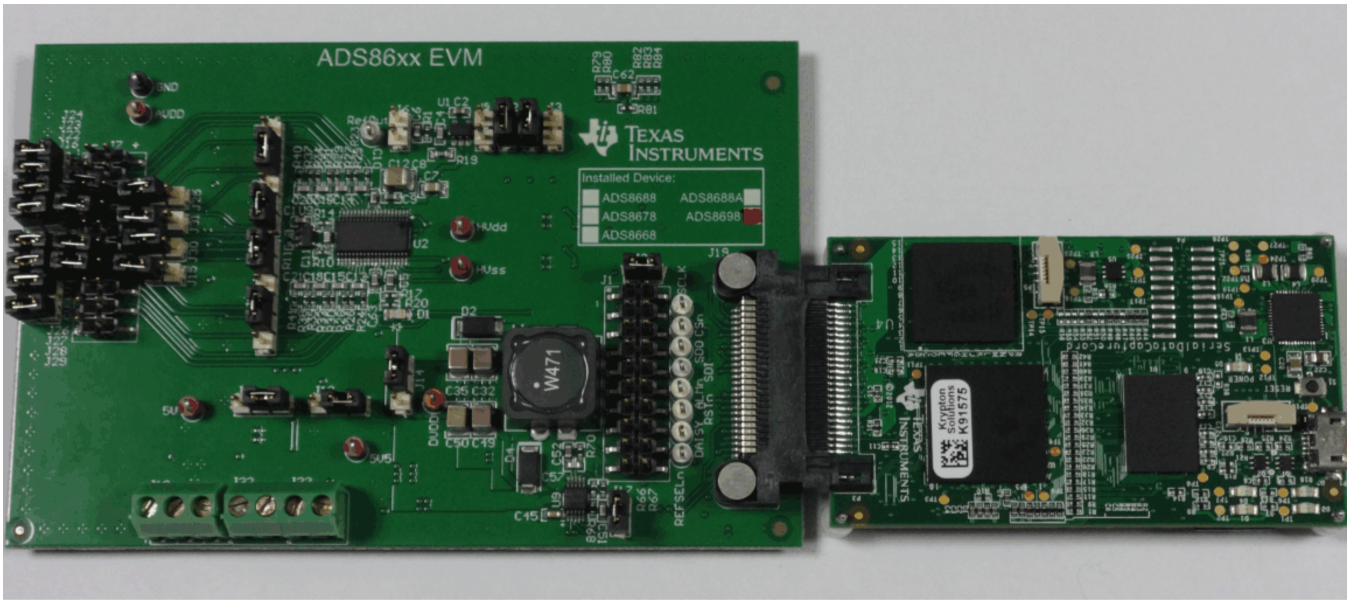
# ADS86xxEVM-PDK Evaluation Module




**ADS86xxEVM-PDK**

This user guide describes the operation and usage of the ADS86xx evaluation module (EVM). The ADS86xx are a family of pin-compatible successive approximation (SAR) analog-to-digital converters (ADCs) with varying resolution and channel counts designed for integrated data acquisition systems. Table 1 lists the devices available in the ADS86xx family that are supported by the ADS86xxEVM.

**Table 1. Devices Supported By the ADS86xxEVM**

| Device | Description |
|---|---|
| ADS8698 | 8-channel, integrated, 18-bit SAR ADC with ALARM |
| ADS8688A | 8-channel, integrated, 16-bit SAR ADC with ALARM |
| ADS8678 | 8-channel, integrated, 14-bit SAR ADC with ALARM |
| ADS8668 | 8-channel, integrated, 12-bit SAR ADC with ALARM |

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# 1 Related Documents

Table 2 lists the related documents that are available through the Texas Instruments web site at www.ti.com.

The ADS86xxEVM is designed to demonstrate true performance of these devices. Each input channel on the selected device can support true bipolar input ranges of ±10.24 V, ±5.12 V, and ±2.56 V as well as unipolar input ranges of 0 V to 10.24 V and 0 V to 5.12 V. The ADS8688A, ADS8678, and ADS8668 devices support additional true bipolar input ranges of ±1.28 V and ±0.64 V as well as unipolar input ranges of 0 V to 2.56 V and 0 V to 1.28 V. The input range selection is done by software programming the device internal registers. This selection is independent for each channel. The device offers a 1-MΩ, constant resistive input impedance irrespective of the selected input range. This user guide includes circuit description, a schematic diagram, and a bill of materials for the ADS86xxEVM circuit board.

**Table 2. Related Documentation**

| Device | Literature Number |
|---|---|
| ADS8698 | SBAS686 |
| ADS8688A | SBAS680 |
| ADS8678 | SBAS627 |
| ADS8668 | SBAS492 |
| OPA376 | SBOS406 |
| OPA2209 | SBOS426 |
| OPA320 | SBOS513 |
| REG71055 | SBAS221 |
| TPS7A4901 | SBVS121 |
| TPS54060 | SLVS919 |
| TPS7A3001 | SBVS125 |


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## 2 ADS86xxEVM-PDK Overview

The ADS86xxEVM-PDK is a platform for evaluating the ADS86xx series of devices. Throughout this document, the terms *ADS86xxEVM*, *demonstration kit*, and *evaluation board* are synonyms with the ADS8698EVM-PDK, ADS8688AEVM-PDK, ADS8678EVM-PDK, and ADS8668EVM-PDK. The ADS86xxEVM-PDK consists of an ADS86xxEVM board and a simple capture card. The simple capture card is a field-programmable gate array (FPGA)-based controller card that functions as a serial peripheral interface (SPI™) host and transfers data to the ADS86xxEVM graphical user interface (GUI) via a USB interface. The ADS86xxEVM GUI collects, analyzes, and records data from the ADS86xxEVM board. The ADS86xxEVM GUI is capable of collecting data from the ADS86xxEVM in auto and manual modes, configuring the ADC program registers, and performing FFT analysis of data captured from the ADC.

**ADS86xxEVM Features**:

- Includes support circuitry as a design example to match ADC performance.
- 3.3-V slave SPI.
- Serial interface header for easy connection to the simple capture card.
- Designed for a 5-V analog supply.
- Integrated 4.096-V voltage reference.
- Bipolar (±10.24 V, ±5,12 V, ±2.56 V, ±1.28 V, and ±0.64 V) or unipolar (0 V to 10.24 V, 0 V to 5.12 V, 0 V to 2.56 V, and 0 V to 1.28 V) input ranges for each channel.
- Onboard, second-order, Butterworth, low-pass filters for four channels.
- Onboard regulator for generating a ±15-V bipolar supply for second-order, Butterworth, low-pass filters.
- Accepts a ±100-mV signal on the negative analog inputs (AIN_xGND).

**ADS86xxEVM GUI Features:**

- Captures data from the ADS86xxEVM in auto and manual modes.
- Configures the ADS86xx device program registers.
- Enables and disables channels in auto mode.
- Provides FFT analysis and calculates the SNR, THD, and SINAD ac performance parameters.
- Provides single and multiple graph views for captured data.
- Includes a dc histogram for dc inputs.
- Logs ADC data.


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## 3 EVM Analog Interface

The ADS86xxEVM is designed for easy interfacing with analog sources. The Samtec™ connector provides a convenient 10-pin, dual-row header at J7. Figure 1 and Figure 2 show the ADS86xxEVM analog input connections for channels AIN0 to AIN3 and channels AIN4 to AIN7, respectively. Table 3 lists the analog interface connections for J7.

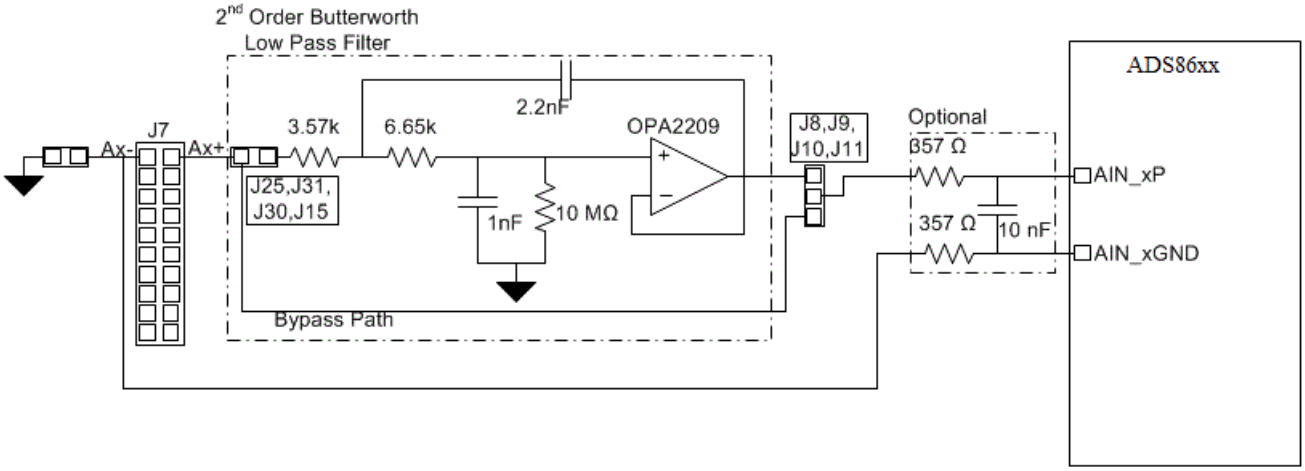


**Figure 1. ADS86xxEVM Analog Input Connections for Channels AIN0, AIN1, AIN2, and AIN3**

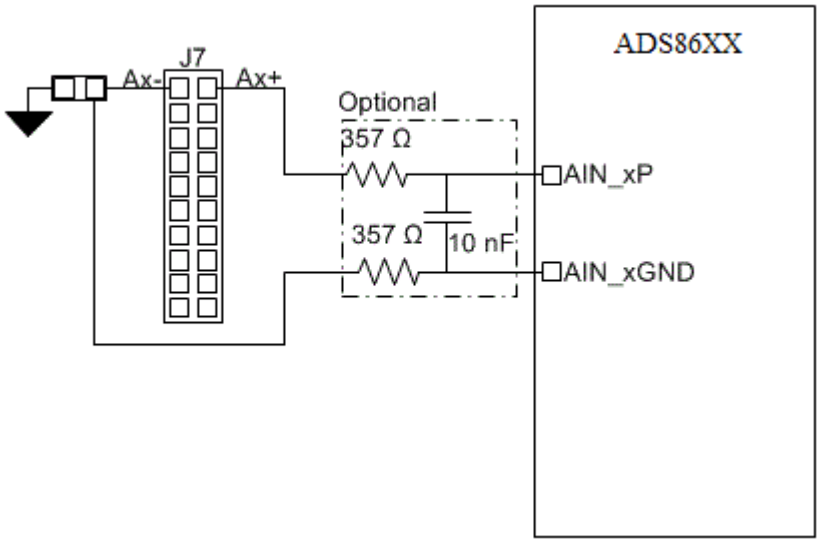


**Figure 2. ADS86xxEVM Analog Input Connections for Channels AIN4, AIN5, AIN6, and AIN7**


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Table 3 summarizes the J7 analog interface connector.

**Table 3. J7: Analog Interface Connections**

| Pin Number | Signal | Description |
|---|---|---|
| J7.2 | A6+ | Positive analog input for channel AIN6 |
| J7.4 | A7+ | Positive analog input for channel AIN7 |
| J7.6 | A0+ | Positive analog input for channel AIN0 |
| J7.8 | A1+ | Positive analog input for channel AIN1 |
| J7.10 | AUX+ | Positive analog input for AUX channel |
| J7.12 | A2+ | Positive analog input for channel AIN2 |
| J7.14 | A3+ | Positive analog input for channel AIN3 |
| J7.16 | A4+ | Positive analog input for channel AIN4 |
| J7.18 | A5+ | Positive analog input for channel AIN5 |
| J7.20 | GND | Analog ground connection |
| J7.1 | A6– | Negative analog input for channel AIN6 |
| J7.3 | A7– | Negative analog input for channel AIN7 |
| J7.5 | A0– | Negative analog input for channel AIN0 |
| J7.7 | A1– | Negative analog input for channel AIN1 |
| J7.9 | AUX– | Connected to analog ground |
| J7.11 | A2– | Negative analog input for channel AIN2 |
| J7.13 | A3– | Negative analog input for channel AIN3 |
| J7.15 | A4– | Negative analog input for channel AIN4 |
| J7.17 | A5– | Negative analog input for channel AIN5 |
| J7.19 | GND | Analog ground connection |

## 3.1 Connecting Negative Inputs to Ground

The negative analog inputs for all channels (except for the AUX channel) are capable of accepting a ±100-mV signal. The negative analog inputs can either be connected to the analog ground or a ±100-mV signal can be applied on these inputs. Table 4 describes the appropriate jumper settings for connecting these inputs to analog ground.

**Table 4. Connecting Negative Analog Inputs to Ground**

| Signal | Jumper | Position for Connecting to Analog Ground | Position for Applying a ±100-mV Signal |
|---|---|---|---|
| A0– | J22 | Closed | Open |
| A1– | J16 | Closed | Open |
| A2– | J26 | Closed | Open |
| A3– | J27 | Closed | Open |
| A4– | J28 | Closed | Open |
| A5– | J29 | Closed | Open |
| A6– | J24 | Closed | Open |
| A7– | J23 | Closed | Open |
| AUX– | NA | Always connected to GND | NA |


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### 3.2 Using Onboard, Second-Order, Butterworth, Low-Pass Filters

The ADS86xxEVM includes second-order, Butterworth, low-pass filters with a cutoff frequency of 22 kHz for channels AIN0, AIN1, AIN2, and AIN3. There is also a provision to bypass these filters. See Figure 1 for an analog input circuit for channels AIN0, AIN1, AIN2, and AIN3. Table 5 lists the jumper settings for using onboard, second-order, Butterworth, low-pass filters and Table 6 lists the jumper settings for bypassing these filters.

**Table 5. Using Onboard, Second-Order, Butterworth, Low-Pass Filters**

| Channel | Jumper | Position | Jumper | Position |
|---|---|---|---|---|
| AIN0 | J25 | Closed | J8 | Closed between pins 1 and 2 |
| AIN1 | J31 | Closed | J10 | Closed between pins 1 and 2 |
| AIN2 | J30 | Closed | J11 | Closed between pins 2 and 3 |
| AIN3 | J15 | Closed | J9 | Closed between pins 2 and 3 |

**Table 6. Bypassing the Onboard, Second-Order, Butterworth, Low-Pass Filters**

| Channel | Jumper | Position | Jumper | Position |
|---|---|---|---|---|
| AIN0 | J25 | Open | J8 | Closed between pins 2 and 3 |
| AIN1 | J31 | Open | J10 | Closed between pins 2 and 3 |
| AIN2 | J30 | Open | J11 | Closed between pins 1 and 2 |
| AIN3 | J15 | Open | J9 | Closed between pins 1 and 2 |

### 3.3 Selecting the Reference Mode for the ADS86xxEVM

The ADS86xxEVM can either operate on an internal or external reference. Table 7 lists the jumper settings for selecting the reference. Table 8 describes the connections for the external reference.

**Table 7. Selecting the Reference for the ADS86xxEVM**

| Jumper | Position for Using Internal Reference | Position for Using External Reference |
|---|---|---|
| J2 | Closed | Open |

**Table 8. External Reference Connections**

| Pin Number | Signal | Description |
|---|---|---|
| J5.1 | REFIN | Input for external reference |
| J5.2 | GND | Analog ground connection |


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# 4 Digital Interface

Connector J19 (Samtec part number ERF8-025-01-L-D-RZ-L-TR socket strip connector) provides the digital I/O connections between the ADS86xxEVM board and the simple capture card. Consult Samtec at www.samtec.com or call 1-800-SAMTEC-9 for a variety of mating connector options. Table 9 summarizes the pin outs for connector J19.

**Table 9. Connector J19 Pin Out**

| Pin Number | Signal | Description |
|---|---|---|
| J19.1 | DAISY | Daisy input for the ADC |
| J19.4 | $\overline{\text{EVM_PRESENT}}$ | EVM present, active low (connected to GND) |
| J19.5 | $\overline{\text{REFSEL}}$ | Reference selection input for the ADC |
| J19.6 | $\overline{\text{RST/PD}}$ | Reset or power-down input for the ADC |
| J19.8 | $\overline{\text{A}}$ | No connection |
| J19.11 | EVM_ID_SDA | $I^2C$ data for the onboard EEPROM |
| J19.12 | EVM_ID_SCL | $I^2C$ clock for the onboard EEPROM |
| J19.13 | 3V3_SDCC | 3.3-V digital supply from the simple capture card |
| J19.14 | 5V_SDCC | Unregulated 5-V supply from the simple capture card |
| J19.33, J19.34 | SCLK | Clock input for the ADC |
| J19.35 | $\overline{\text{CS}}$ | Chip-select input for the ADC |
| J19.38 | SDI | Data input for the ADC |
| J19.39 | SDO | Data output from the ADC |
| J19.45-49 | EVMSDxxxxx | Digital connections for the onboard micro secure digital (microSD) card |
| J19.2, J19.10, J19.16, J19.50 | GND | Ground connections |

## 4.1 Serial Interface (SPI)

The ADS86xx device uses SPI serial communication in mode 1 (CPOL = 0, CPHA = 1) with clock speeds up to 18 MHz. The ADS86xxEVM offers 49.9-Ω resistors between the SPI signals and J19 to aid with signal integrity. Typically, in high-speed SPI communication, fast signal edges can cause overshoot; these 49.9-Ω resistors slow down the signal edges in order to minimize signal overshoot.

## 4.2 $I^2C$ Bus for the Onboard EEPROM

The ADS86xxEVM has an $I^2C$ bus that records the board name and assembly date to communicate with the onboard EEPROM. The bus is not used in any form by the ADS86xx converter.

## 4.3 microSD Card

The ADS86xxEVM has a microSD card that contains the software files for the simple capture card. The contents of the microSD card must not be deleted or altered.


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# 5 Power Supplies

The ADS86xxEVM can be powered from the simple capture card if onboard buffers for active low-pass filters are not being used. The onboard regulator (U9) for generating high-voltage supplies (HVDD and HVSS) is disabled by closing jumper J17. The HVDD and HVSS supplies are only required for buffers U4 and U5.

**CAUTION**

Do not open jumper J17 if the ADS86xxEVM must be powered only from the simple capture card and an external 5-V supply is not provided on J32.

High-voltage supplies (HVDD and HVSS) for buffers U4 and U5 can be generated using the onboard regulator (U9) if an external 5-V dc supply is provided on J32. The external 5-V dc supply must be at least 200 mV above the unregulated 5-V supply of the simple capture card. The external 5-V dc supply must be capable of providing at least 500 mA of current. Table 10 provides jumper settings for generating HVDD and HVSS using the onboard switching regulator U9.

**Table 10. Jumper Settings for Generating HVDD and HVSS Using an Onboard Switching Regulator**

| Jumper | Position for Using an Onboard Switching Regulator |
|---|---|
| J12 | Closed between pins 1 and 2 |
| J14 | Closed between pins 1 and 2 |
| J17 | Open |

HVDD and HVSS for buffers U4 and U5 can also be generated by providing external high-voltage supplies on J18, as shown in Table 11. Table 12 and Figure 3 illustrate the power-supply connections for external supplies.

**Table 11. Jumper Settings for Generating HVDD and HVSS from External High-Voltage Supplies**

| Jumper | Position for Generating HVDD and HVSS from External Voltage Supplies |
|---|---|
| J12 | Closed between pins 2 and 3 |
| J14 | Closed between pins 2 and 3 |
| J17 | Closed |

**Table 12. Power-Supply Connections**

| Voltage Supply | Signal | Voltage Range | Pin Number | Note |
|---|---|---|---|---|
| External 5 V | EXT_5V | 5 V to 5.5 V | J32.2 | Required only for generating HVDD and HVSS using the onboard switching regulator |
| — | GND | GND | J32.1 | — |
| External HVDD | EXT_HVDD | 16 V to 25 V | J18.3 | Required only for generating HVDD and HVSS from external high-voltage supplies. Make sure HVDD + \|HVSS\| < 36 V. |
| External HVSS | EXT_HVSS | –16 V to –25 V | J18.1 | Required only for generating HVDD and HVSS from external high-voltage supplies. Make sure HVDD + \|HVSS\| < 36 V. |
| — | GND | GND | J18.2 | — |


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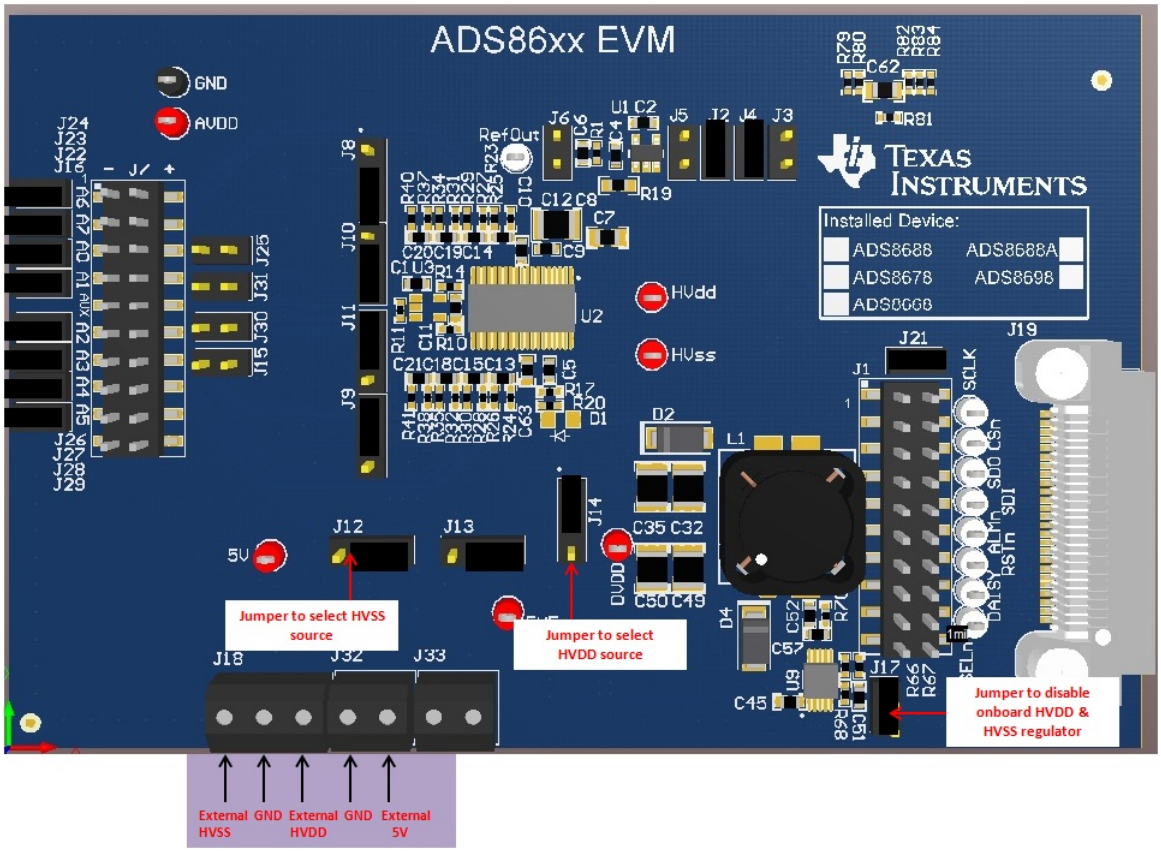


**Figure 3. Power-Supply Connections Diagram**

The AVDD analog supply for the ADS86xx is generated by converting an unregulated 5-V supply from the simple capture card or by converting an external 5-V supply to a regulated 5-V supply by using the REG71055 charge pump and the TPS7A4901 linear regulator. The DVDD digital supply for the ADC is derived from a 3.3-V supply from the simple capture card.


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# 6 ADS86xxEVM-PDK Initial Setup

This section presents the steps required to setup the ADS86xxEVM-PDK kit prior to operation.

## 6.1 Default Jumper Settings

Figure 4 details the default jumper settings. Table 13 provides the configuration for these jumpers.

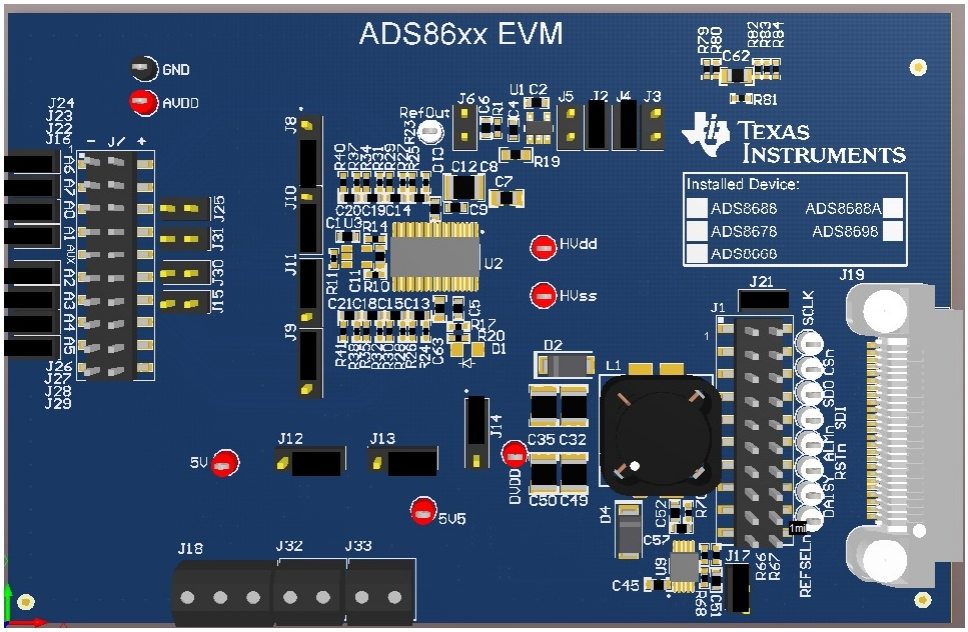


**Figure 4. ADS86xxEVM Default Jumper Settings**



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**Table 13. Default Jumper Configuration**

| Jumper | Default Position |
|---|---|
| J2 | Closed |
| J3 | Open |
| J4 | Closed |
| J5 | Open |
| J6 | Open |
| J8 | Closed between pins 2 and 3 |
| J9 | Closed between pins 1 and 2 |
| J10 | Closed between pins 2 and 3 |
| J11 | Closed between pins 1 and 2 |
| J12 | Closed between pins 1 and 2 |
| J13 | Closed between pins 2 and 3 |
| J14 | Closed between pins 1 and 2 |
| J15 | Open |
| J16 | Closed |
| J17 | Closed |
| J21 | Closed |
| J22 | Closed |
| J23 | Closed |
| J24 | Closed |
| J25 | Open |
| J26 | Closed |
| J27 | Closed |
| J28 | Closed |
| J29 | Closed |
| J30 | Open |
| J31 | Open |

## 6.2 Software Installation

This section presents the steps required to the install the software.

**NOTE:** Ensure the microSD memory card included in the kit is installed in the microSD socket (P6) on the back of the simple capture card before connecting the EVM to the computer. Otherwise, as a result of improper boot up, Windows® cannot recognize the ADS86xxEVM-PDK as a connected device.

Complete the following steps to install the software:

1. Verify the microSD memory cards are installed on the simple capture card and the ADS86xxEVM board.
2. Verify jumpers are in the factory-default position and are properly connected to the hardware.
3. Install the ADS86xxEVM-PDK software.
4. Complete the *simple capture card* device driver installation.

Each task is described in the following subsections.


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### 6.2.1 Verify the microSD Memory Card is Installed on the Simple Capture Card

The ADS86xxEVM-PDK includes microSD memory cards that contain the EVM software and simple capture card firmware required for the EVM operation.

**NOTE:** Ensure the microSD memory cards that contain the software are installed in the microSD socket on the back of the simple capture card and on the back of ADS86xxEVM board. Figure 5 and Figure 6 show the bottom view of the simple capture card and the ADS86xxEVM, respectively, with the microSD card installed.

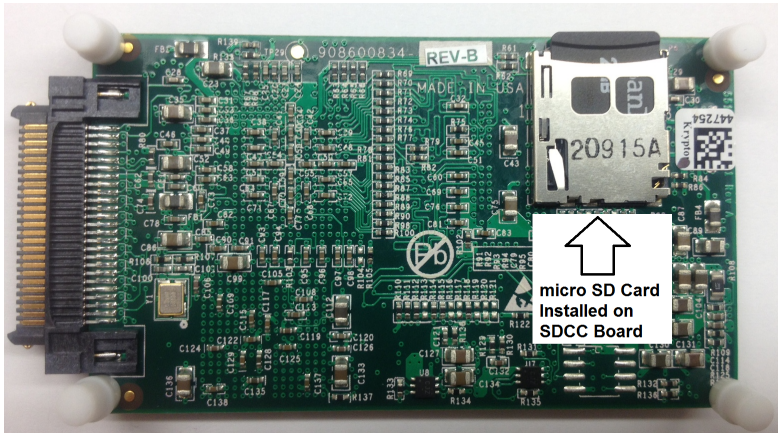


**Figure 5. Bottom View of the Simple Capture Card with the microSD Memory Card Installed**

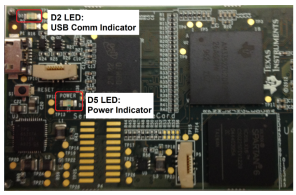


**Figure 6. Bottom View of the ADS86xxEVM Board with the microSD Memory Card Installed**

The microSD memory cards are formatted at the factory with the necessary firmware files for the simple capture card to boot properly. In addition to the simple capture card firmware files (application and MLO files), the microSD memory cards contain the ADS86xxEVM-PDK software installation files.

### 6.2.2 Verify Jumpers are in the Factory-Default Position and Connect the Hardware

The ADS86xxEVM-PDK includes both the ADS86xxEVM and the simple capture card; however, the devices are shipped unconnected. Follow these steps to verify the configuration and connectivity of the ADS86xxEVM-PDK.

1. Ensure that he microSD card is installed on the back of the Simple Capture card and ADS86xxEVM respectively, as shown in Figure 5 and Figure 6.
2. Ensure that the ADS86xxEVM jumpers are configured as illustrated in Figure 4.
3. Connect the ADS86xxEVM board to the simple capture card; see Figure 7.



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### 7.5.2 FFT Analysis

The FFT analysis page in the GUI performs the fast fourier transform (FFT) of the captured data and displays the resulting frequency domain plots. This page also calculates key ADC dynamic performance parameters, such as signal-to-noise ratio (SNR), total harmonic distortion (THD), signal-to-noise and distortion ratio (SINAD), and spurious-free dynamic range (SFDR). Figure 26 shows the FFT performance analysis display. The input channel (AINx) for FFT analysis can be selected from the channel drop-down menu and the data capture settings can be entered in boxes on the left side of the graph. The FFT calculated parameters are shown on the bottom side of the graph.

**Figure 26. FFT Analysis**

Input signal frequency for FFT analysis:

- The GUI analyses the FFT to provide signal power (dBFs), SNR, SINAD, THD, SFDR, and ENOB numbers.
- The GUI uses a 7-term Blackman-Harris window to minimize spectral leakage.


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## 7.6 Alarm Feature

### 7.6.1 Alarm Functionality

The ADS86xx series of devices support an elaborate set of ALARM functionality. The detail of the ALARM functionality and operation is available in respective device data sheets. The ADS86xxEVM and GUI support the required hardware and software features to demonstrate the ALARM functionality of the device.

The ALARM feature of the device can be activated by clicking the **Edit Alarm Limits** button when in manual or auto mode. Individual channel threshold levels and hysteresis levels can be programmed in the *Configure Alarm Settings* pop-up window. The ALARM feature can be enabled or disabled by checking or unchecking the **Alarm Enable?** check box. Click the **Config Alarm Settings** button to program device registers with correct data, as shown in Figure 27.

The user can also program these values by individually programming each register from register map view, as described in *Section 7.3.2*.

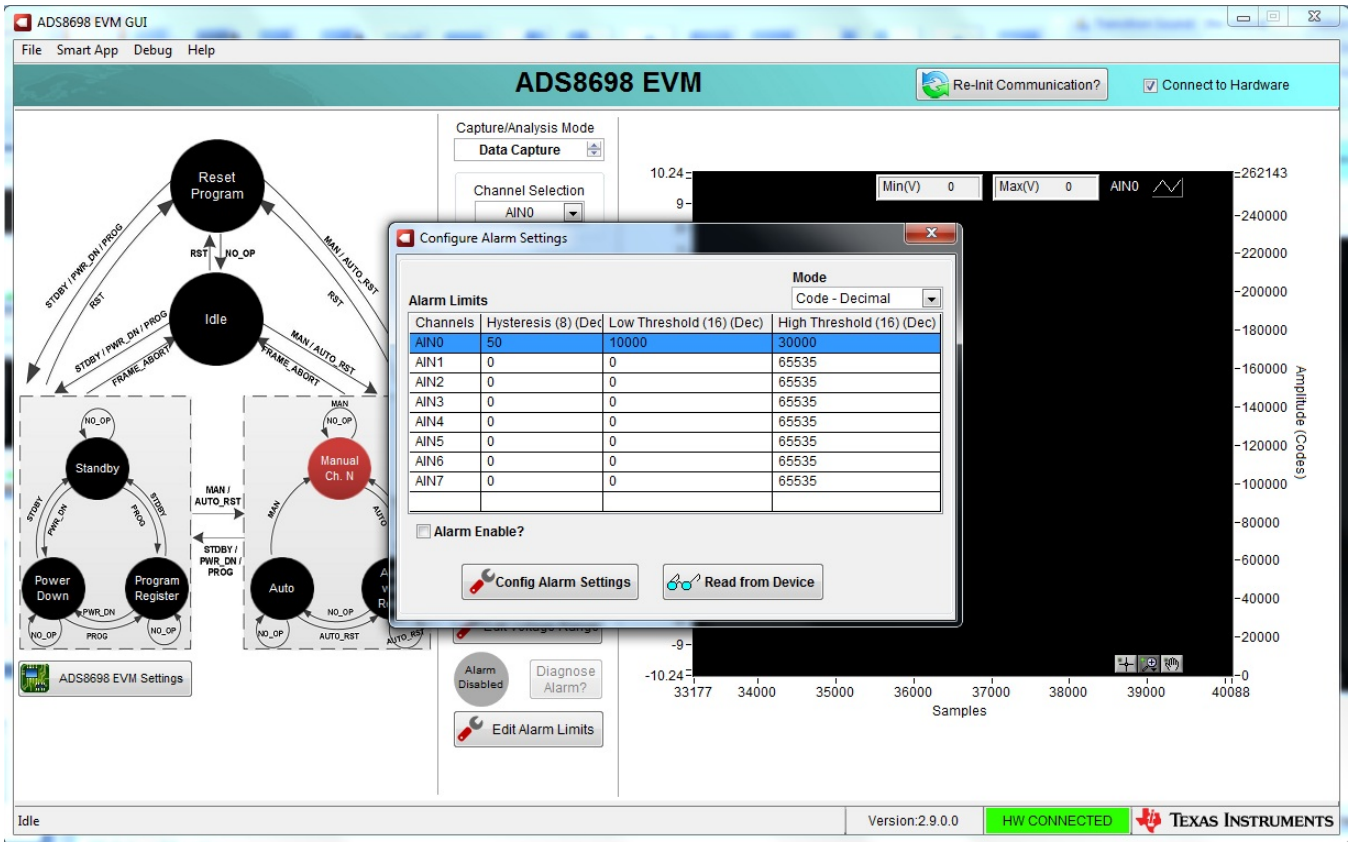


**Figure 27. Alarm Enable Settings**



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When the ALARM feature is enabled, the capture data window shows the indicators for high and low thresholds based on the values programmed. After data capture, if the device detects an ALARM event, EVM diode D1 turns on with a red indicator light. The GUI monitors the status of the ALARM pin on the device. When an ALARM is detected, the ALARM indicator on the GUI starts flashing.

The user can read the status of tripped alarms by clicking the **Diagnose Alarm?** button. The user can also read the ALARM register contents from register map view to detect the state of tripped and active alarms.

## 7.7 Phase Compensation

The ADS86xxEVM GUI includes an analysis page for compensating the phase of signals captured in auto mode. When the signals on different channels are sampled in auto mode, a deterministic phase difference between signals is introduced resulting from the time difference between sampling instants. The phase difference is dependent on sampling rate, input signal frequency, number of channels, and initial phase difference. The phase compensation analysis page compensates for the introduced phase difference and provides the results after phase compensation. For details on phase compensation, see the reference guide *Phase Compensated 8-CH Multiplexed Data Acquisition System for Power Automation* (TIDU427). The phase compensation analysis page can be activated from the *Smart App* menu. Figure 28 displays the phase compensation analysis page.


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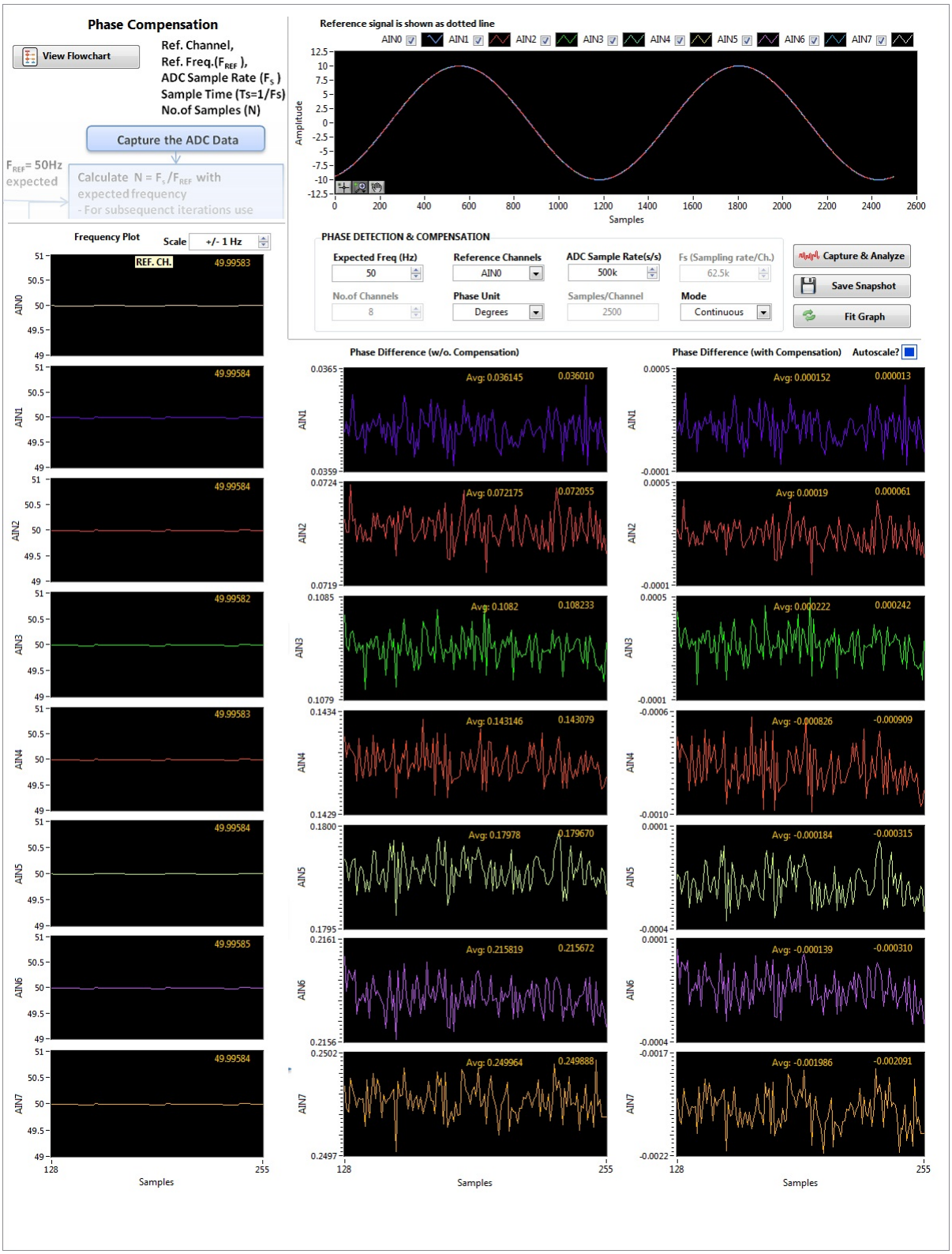


**Figure 28. Phase Compensation Analysis**



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## 7.8 ADS86xxEVM GUI Simulation Mode

The ADS86xxEVM GUI can be run in simulation mode by unchecking the *Connect to Hardware* check-box on the top right side of the GUI window. In simulation mode, the GUI does not connect to the ADS86xxEVM board and only displays the results for one set of captured data stored in the computer. Figure 29 shows the ADS86xxEVM GUI running in simulation mode.

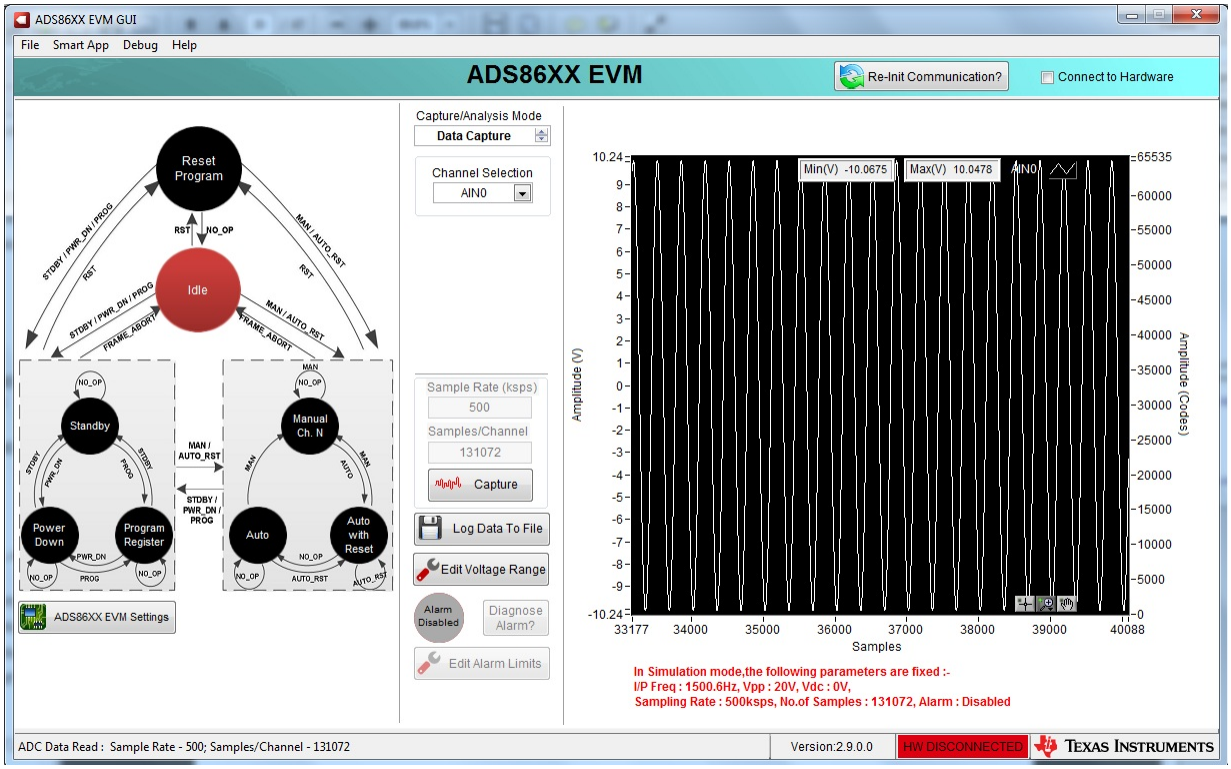


**Figure 29. Simulation Mode**


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# 8 Bill of Materials, Layout, and Schematics

Schematics for the ADS86xxEVM are appended to this user's guide. The bill of material is the same for all devices in the ADS86xx family. The device U2 varies based on the device chosen for evaluation. The bill of materials is provided in Table 14. Section 8.2 shows the PCB layouts for the ADS86xxEVM with the ADS8688IDBT device installed on the EVM. Section 8.3 provides the schematics for the ADS86xxEVM and ADS8688IDBT.

## 8.1 Bill of Materials

**NOTE:** All components are compliant with the European Union Restriction on Use of Hazardous Substances (RoHS) Directive. Some part numbers may be either leaded or RoHS. Verify that purchased components are RoHS-compliant. (For more information about TI's position on RoHS compliance, see www.ti.com.)

**Table 14. ADS8688EVM Bill of Materials**

| Item No. | Qty | Ref Des | Description | Vendor | Part Number |
|---|---|---|---|---|---|
| 1 | 1 | !PCB | Printed Circuit Board | Any | PRJ_Number |
| 2 | 4 | C1-C3, C6 | CAP, CERM, 0.1uF, 25V, +/-5%, X7R, 0603 | AVX | 06033C104JAT2A |
| 3 | 1 | C10 | CAP, CERM, 1uF, 16V, +/-10%, X5R, 0603 | Kemet | C0603C105K4PACTU |
| 4 | 1 | C11 | CAP, CERM, 3000pF, 50V, +/-5%, C0G/NP0, 0603 | MuRata | GRM1885C1H302JA01D |
| 5 | 6 | C16, C24, C26, C30, C45, C64 | CAP, CERM, 0.1uF, 50V, +/-10%, X7R, 0603 | AVX | 06035C104KAT2A |
| 6 | 4 | C17, C22, C27, C28 | CAP, CERM, 2200pF, 50V, +/-5%, C0G/NP0, 0603 | TDK | C1608C0G1H222J |
| 7 | 4 | C23, C25, C29, C31 | CAP, CERM, 1000pF, 50V, +/-5%, C0G/NP0, 0402 | MuRata | GRM1555C1H102JA01D |
| 8 | 4 | C32, C35, C49, C50 | CAP, CERM, 22uF, 25V, +/-10%, X7R, 1210 | MuRata | GRM32ER71E226KE15L |
| 9 | 3 | C34, C36, C53 | CAP, CERM, 10uF, 25V, +/-10%, X7R, 1206 | MuRata | GRM31CR71E106KA12L |
| 10 | 2 | C37, C54 | CAP, CERM, 10uF, 35V, +/-10%, X7R, 1206 | Taiyo Yuden | GMK316AB7106KL |
| 11 | 15 | C4, C12-C15, C18-C21, C38-C41, C56, C58 | CAP, 10000pF, 0603, 5%, 50V, C0G | TDK | C1608C0G1H103J080AA |
| 12 | 1 | C42 | CAP, CERM, 0.22uF, 16V, +/-10%, X7R, 0603 | TDK | C1608X7R1C224K |
| 13 | 2 | C44, C46 | CAP, CERM, 2.2uF, 10V, +/-10%, X7R, 0603 | MuRata | GRM188R71A225KE15D |
| 14 | 1 | C48 | CAP, CERM, 10uF, 50V, +/-20%, X7R, 2220 | TDK | C5750X7R1H106M |
| 15 | 1 | C5 | CAP, CERM, 10uF, 6.3V, +/-20%, X5R, 0603 | Kemet | C0603C106M9PACTU |
| 16 | 1 | C51 | CAP, CERM, 0.1uF, 25V, +/-5%, X7R, 0603 | Kemet | C0603C104J3RAC |
| 17 | 1 | C52 | CAP, CERM, 0.39uF, 16V, +/-10%, X7R, 0603 | MuRata | GRM188R71C394KA88D |
| 18 | 1 | C57 | CAP, CERM, 470pF, 50V, +/-10%, X7R, 0603 | TDK | C1608X7R1H471K |
| 19 | 2 | C59, C60 | CAP, CERM, 10uF, 35V, +/-10%, X7R, 1210 | MuRata | GRM32ER7YA106KA12L |
| 20 | 7 | C7, C33, C43, C47, C55, C61, C62 | CAP, CERM, 10uF, 10V, +/-10%, X7R, 0805 | MuRata | GRM21BR71A106KE51L |
| 21 | 1 | C8 | CAP, CERM, 22uF, 16V, +/-20%, X7R, 1210 | TDK | C3225X7R1C226M |
| 22 | 2 | C9, C63 | CAP, CERM, 1uF, 16V, +/-10%, X7R, 0603 | TDK | C1608X7R1C105K |
| 23 | 1 | D1 | Diode, LED, RED, 2.1V, 14.2-mcd, 20 mA, 0805 | Lite On | LTST-C170UKT |



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**Table 14. ADS8688EVM Bill of Materials (continued)**

| Item No. | Qty | Ref Des | Description | Vendor | Part Number |
|---|---|---|---|---|---|
| 24 | 2 | D2, D4 | Diode, Schottky, 60V, 2A, SMA | Diodes Inc. | B260A-13-F |
| 25 | 2 | D3, D5 | Diode, Zener, 5.6V, 500mW, SOD-123 | ON Semiconductor | MMSZ4690T1G |
| 26 | 2 | D6, D8 | Diode, Zener, 27V, 500mW, SOD-123 | Vishay-Semiconductor | MMSZ4711-V |
| 27 | 1 | D7 | Diode, Zener, 3.9V, 500mW, SOD-123 | ON Semiconductor | MMSZ4686T1G |
| 28 | 2 | D9, D10 | Diode, Schottky, 30V, 0.2A, SOT-23 | Diodes Inc. | BAT54C-7-F |
| 29 | 6 | FID1-FID6 | Fiducial mark. There is nothing to buy or mount. | N/A | N/A |
| 30 | 4 | H1-H4 | Bumpon, Hex, 0.063mil, 11mm Dia, Lt Brn | 3M | SJ5202 |
| 31 | 2 | J1, J7 | Header, 100mil, 10x2, SMD | Samtec, Inc. | TSM-110-01-T-DV-P |
| 32 | 1 | J18 | Terminal Block, 6A, 3.5mm Pitch, 3-Pos, TH | On-Shore Technology, Inc. | ED555/3DS |
| 33 | 1 | J19 | Receptacle, Micro High Speed Socket Strip, 0.8mm, 25x2, R/A, SMT | Samtec, Inc. | ERF8-025-01-L-D-RA-L-TR |
| 34 | 19 | J2-J6, J15-J17, J21-J31 | Header, TH, 100mil, 2x1, Gold plated, 230 mil above insulator | Samtec, Inc. | TSW-102-07-G-S |
| 35 | 1 | J20 | MicroSD Memory Card Connector | Molex | 502570-0893 |
| 36 | 2 | J32, J33 | Terminal Block, 6A, 3.5mm Pitch, 2-Pos, TH | On-Shore Technology, Inc. | ED555/2DS |
| 37 | 7 | J8-J14 | Header, TH, 100mil, 3x1, Gold plated, 230 mil above insulator | Samtec, Inc. | TSW-103-07-G-S |
| 38 | 1 | L1 | — | Wurth Electronics | 744870471 |
| 39 | 1 | Q1 | Transistor, NPN, 45V, 0.1A, SOT-23 | ON Semiconductor | BC847CLT1G |
| 40 | 8 | R1, R79-R85 | RES, 10.0k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW040210K0FKED |
| 41 | 5 | R11, R21, R22, R74, R76 | RES, 10.0Meg ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW040210M0FKED |
| 42 | 1 | R12 | RES, 1.0k ohm, 5%, 0.125W, 0805 | Vishay-Dale | CRCW08051K00JNEA |
| 43 | 3 | R13, R15, R18 | RES, 47.0k ohm, 1%, 0.1W, 0603 | Yageo America | RC0603FR-0747KL |
| 44 | 1 | R17 | RES, 220k ohm, 5%, 0.063W, 0402 | Vishay-Dale | CRCW0402220KJNED |
| 45 | 1 | R19 | RES, 100 ohm, 1%, 0.1W, 0603 | Vishay-Dale | CRCW0603100RFKEA |
| 46 | 17 | R2, R3, R5-R10, R14, R16, R86-R91, R93 | RES, 49.9 ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW040249R9FKED |
| 47 | 1 | R20 | RES, 100k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW0402100KFKED |
| 48 | 16 | R23-R32, R34, R35, R37, R38, R40, R41 | RES, 357 ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW0402357RFKED |
| 49 | 13 | R36, R44, R48, R52, R94-R96, R98-R100, R102, R103, R107 | Resistor, Uninstalled | NI | NI |
| 50 | 16 | R4, R33, R39, R47, R49, R61, R69, R77, R78, R92, R97, R101, R104-R106, R108 | RES, 0 ohm, 5%, 0.063W, 0402 | Yageo America | RC0402JR-070RL |
| 51 | 4 | R42, R46, R50, R54 | RES, 3.57k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW04023K57FKED |
| 52 | 4 | R43, R45, R51, R53 | RES, 6.65k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW04026K65FKED |
| 53 | 1 | R55 | RES, 576k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW0402576KFKED |
| 54 | 2 | R56, R71 | RES, 232k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW0402232KFKED |
| 55 | 1 | R57 | RES, 182k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW0402182KFKED |
| 56 | 4 | R58, R60, R62, R72 | RES, 499k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW0402499KFKED |
| 57 | 2 | R59, R73 | RES, 20k ohm, 5%, 0.063W, 0402 | Vishay-Dale | CRCW040220K0JNED |
| 58 | 1 | R63 | RES, 33k ohm, 5%, 0.063W, 0402 | Vishay-Dale | CRCW040233K0JNED |



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**Table 14. ADS8688EVM Bill of Materials (continued)**

| Item No. | Qty | Ref Des | Description | Vendor | Part Number |
|---|---|---|---|---|---|
| 59 | 2 | R64, R75 | RES, 82k ohm, 5%, 0.063W, 0402 | Vishay-Dale | CRCW040282K0JNED |
| 60 | 1 | R65 | RES, 15k ohm, 5%, 0.063W, 0402 | Vishay-Dale | CRCW040215K0JNED |
| 61 | 1 | R66 | RES, 1.1k ohm, 5%, 0.063W, 0402 | Vishay-Dale | CRCW04021K10JNED |
| 62 | 1 | R67 | RES, 44.2k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW040244K2FKED |
| 63 | 1 | R68 | RES, 412k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW0402412KFKED |
| 64 | 1 | R70 | RES, 20.5k ohm, 1%, 0.063W, 0402 | Vishay-Dale | CRCW040220K5FKED |
| 65 | 1 | SD1 | SanDisk MicroSD Card, 2GB | SanDisk | SDSDQ-002G |
| 66 | 26 | SH-J2-SH-J6, SH-J8-SH-J17, SH-J21-SH-J31 | Shunt, 100mil, Gold plated, Black | 3M | 969102-0000-DA |
| 67 | 1 | TP11 | Test Point, Miniature, Black, TH | Keystone | 5001 |
| 68 | 6 | TP12-TP17 | Test Point, Miniature, Red, TH | Keystone | 5000 |
| 69 | 9 | TP2-TP10 | Test Point, Miniature, White, TH | Keystone | 5002 |
| 70 | 2 | U1, U3 | Op Amp, Precision, 5.5MHz, Low-Noise, RRIO, with e-trim | Texas Instruments | OPA376AIDBVR |
| 71 | 1 | U10 | IC, -3V to -36V, -200mA, Ultralow Noise, High-PSRR LDO Negative Linear Regulator | Texas Instruments | TPS7A3001DGN |
| 72 | 1 | U11 | IC, 2K, Serial EEPROM | Atmel | AT24C02B |
| 73 | 1 | U2 | 16 bit 500KSPS 8 Channel SAR ADC | Texas Instruments | ADS8688IDBT |
| 74 | 2 | U4, U5 | OpAmp, Low Noise, Low Power, 36V | Texas Instruments | OPA2209AIDGKR |
| 75 | 2 | U6, U7 | IC, VIN 3V to 35V, 150mA, Ultralow Noise, High-PSRR, LDO Regulator | Texas Instruments | TPS7A4901DGN |
| 76 | 1 | U8 | IC, SWITCHED CAP, BUCK BOOST CONVERTER 1.8V to 5.5V in 65uA | Texas Instruments | REG71055DDC |
| 77 | 1 | U9 | Buck Inverting Buck-Boost Step Down Regulator with 3.5 to 60 V Input and 0.8 to 58 V Output, -40 to 150 degC, 10-Pin MSOP-PowerPAD (DGQ), Green (RoHS & no Sb/Br) | Texas Instruments | TPS54060DGQ |



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## 8.2 Board Layouts

Figure 30 through Figure 35 show the printed circuit board (PCB) layouts for the ADS86xxEVM.

**NOTE:** Board layouts are not to scale. These figures are intended to show how the board is laid out; these figures are not intended to be used for manufacturing ADS86xxEVM PCBs.

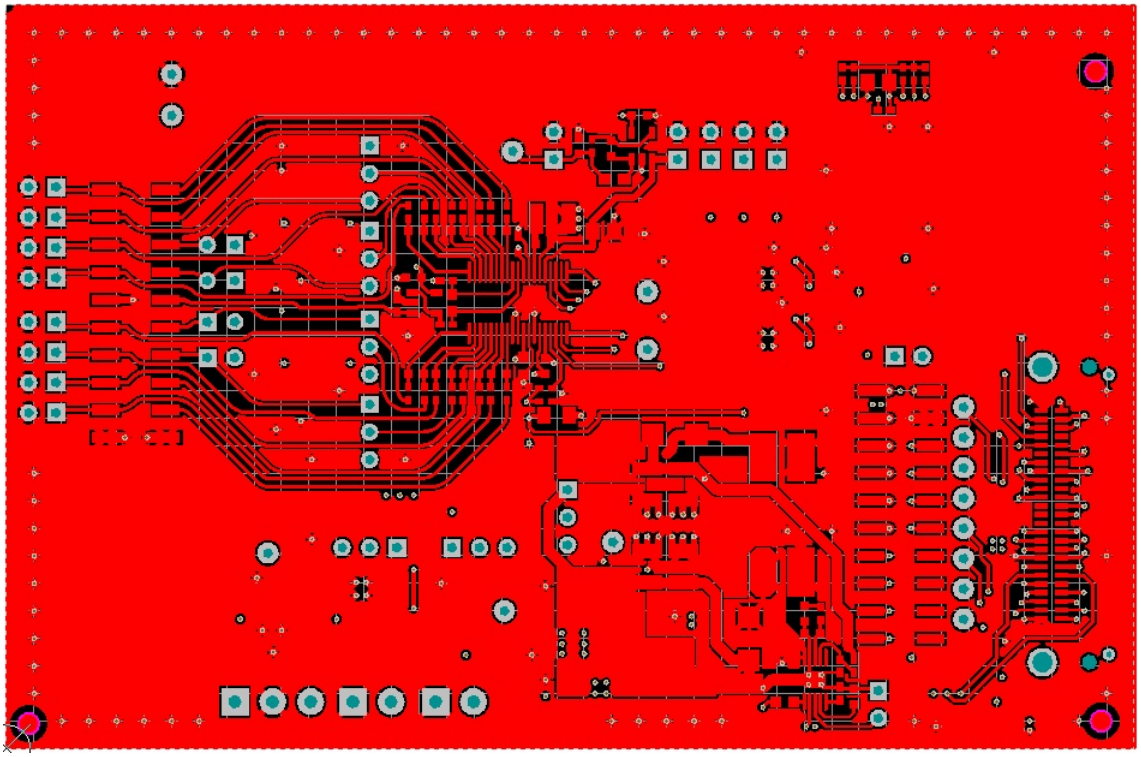

**Figure 30. ADS86xxEVM PCB: Top Layer (L1)**


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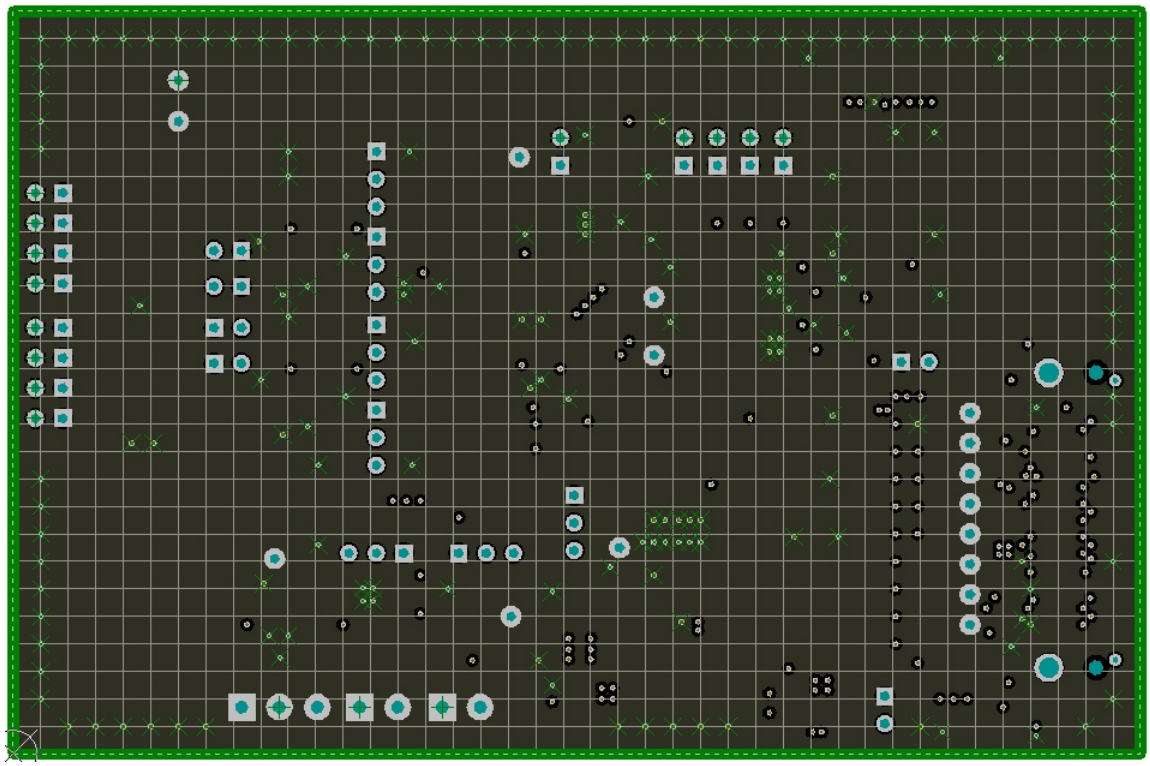

**Figure 31. ADS86xxEVM PCB: Ground Layer (L2)**

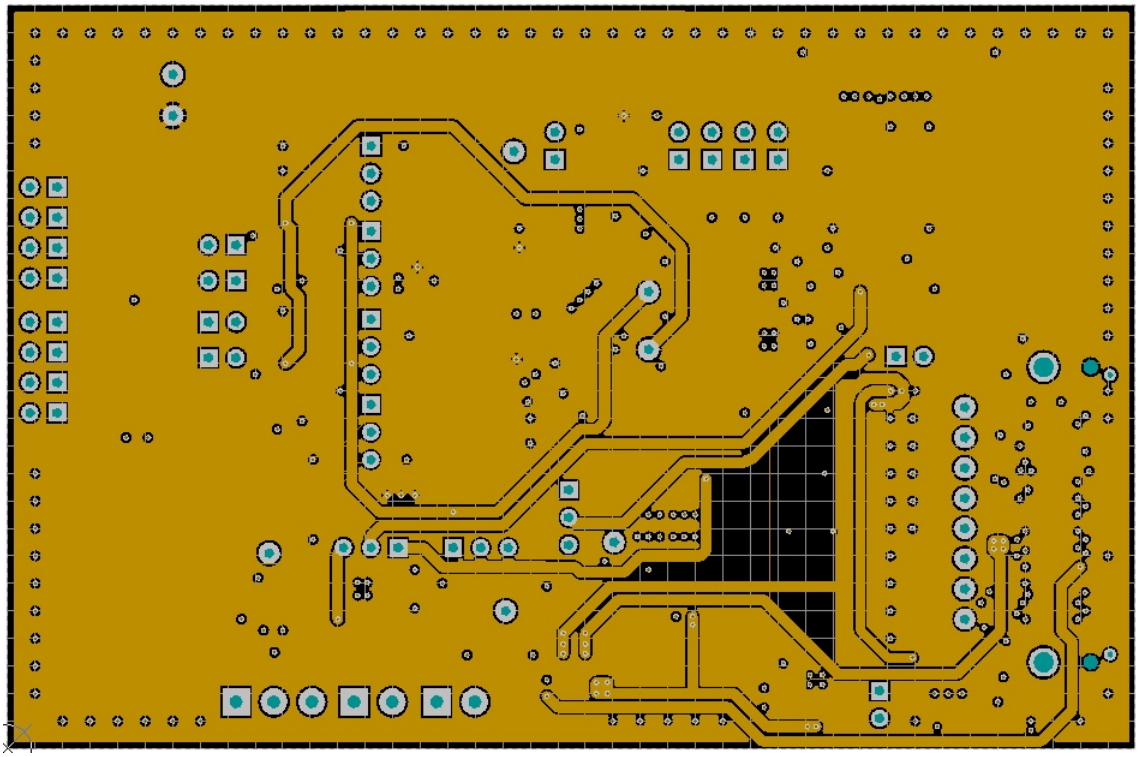

**Figure 32. ADS86xxEVM PCB: Analog Power Layer (L3)**



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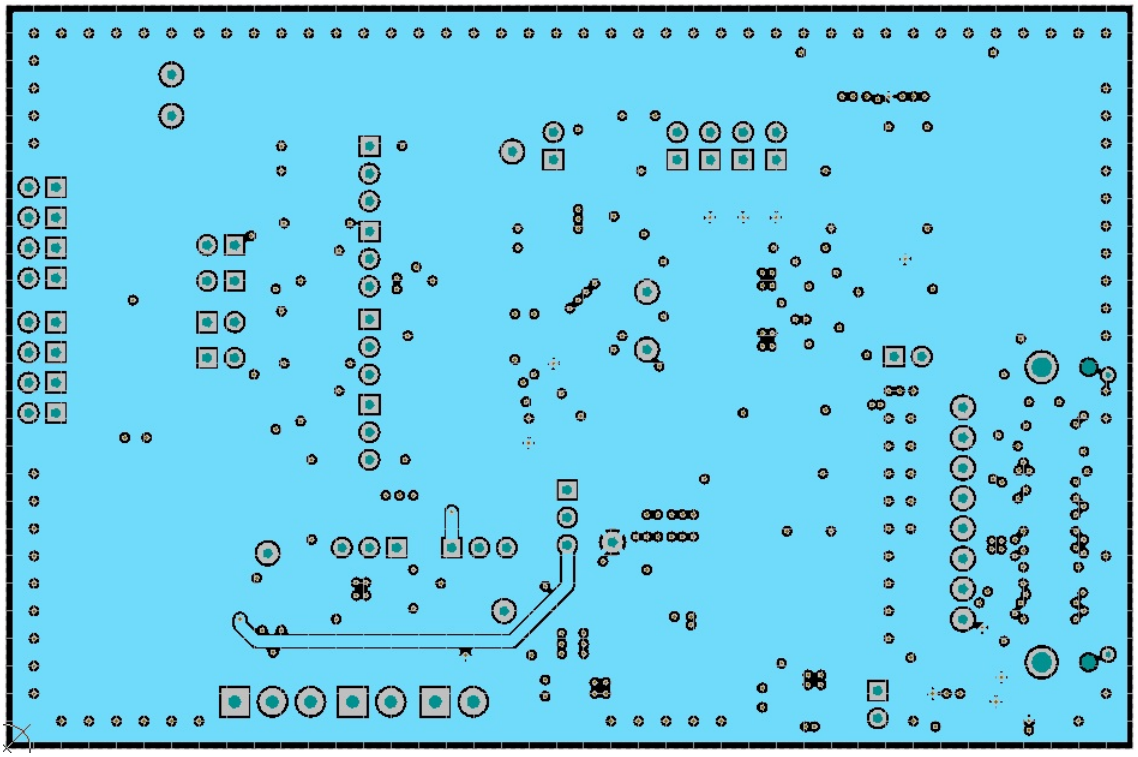

**Figure 33. ADS86xxEVM PCB: Digital Power Layer (L4)**

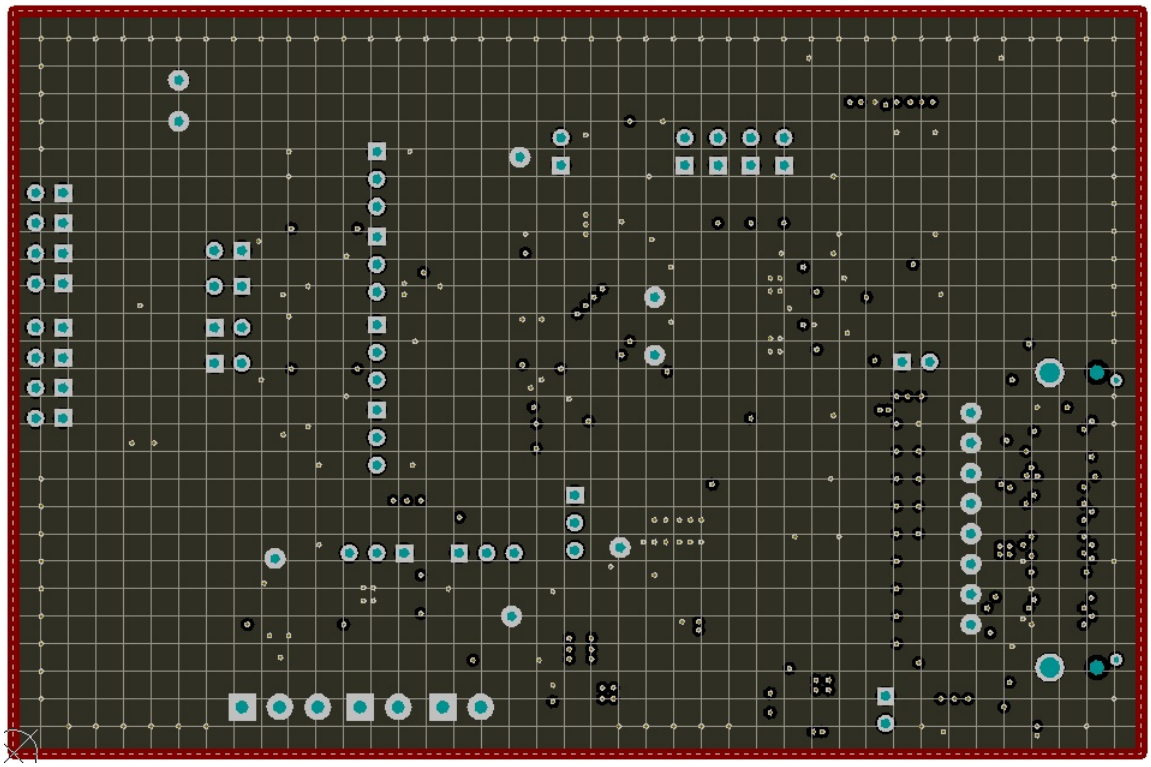

**Figure 34. ADS86xxEVM PCB: Ground Layer (L5)**


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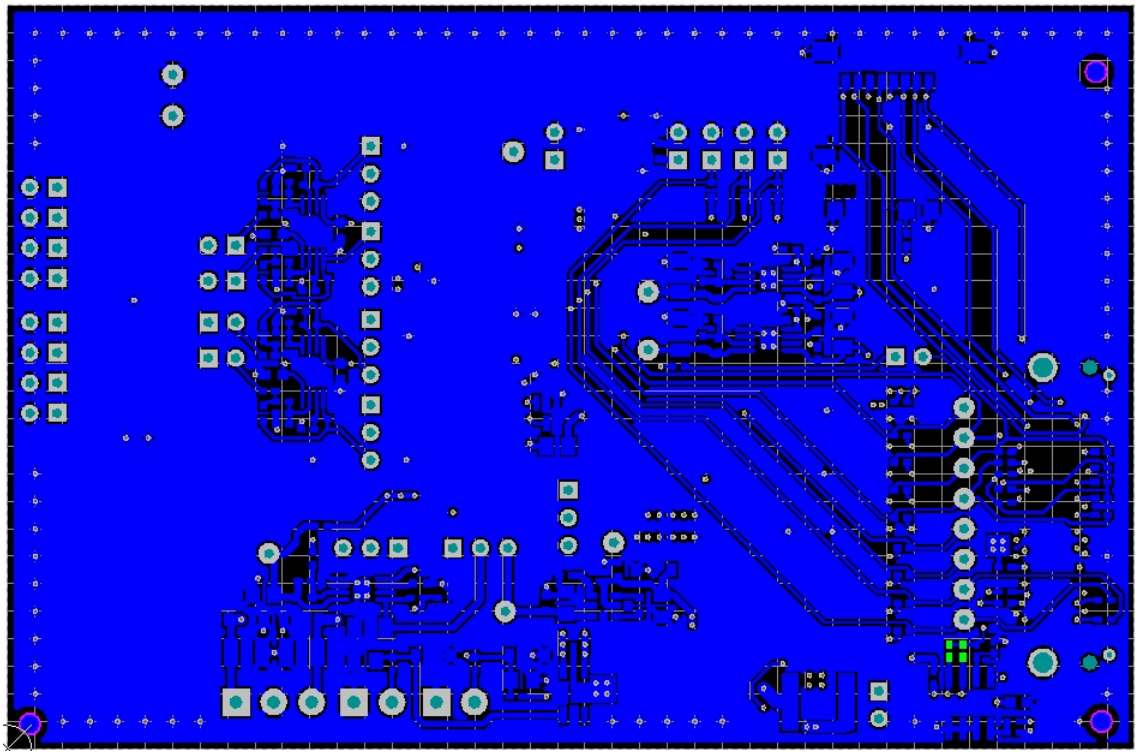

**Figure 35. ADS86xxEVM PCB: Bottom Layer (L6)**



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## 8.3 Schematics

Figure 36 and Figure 37 illustrate schematics for the ADS86xxEVM-PDK.

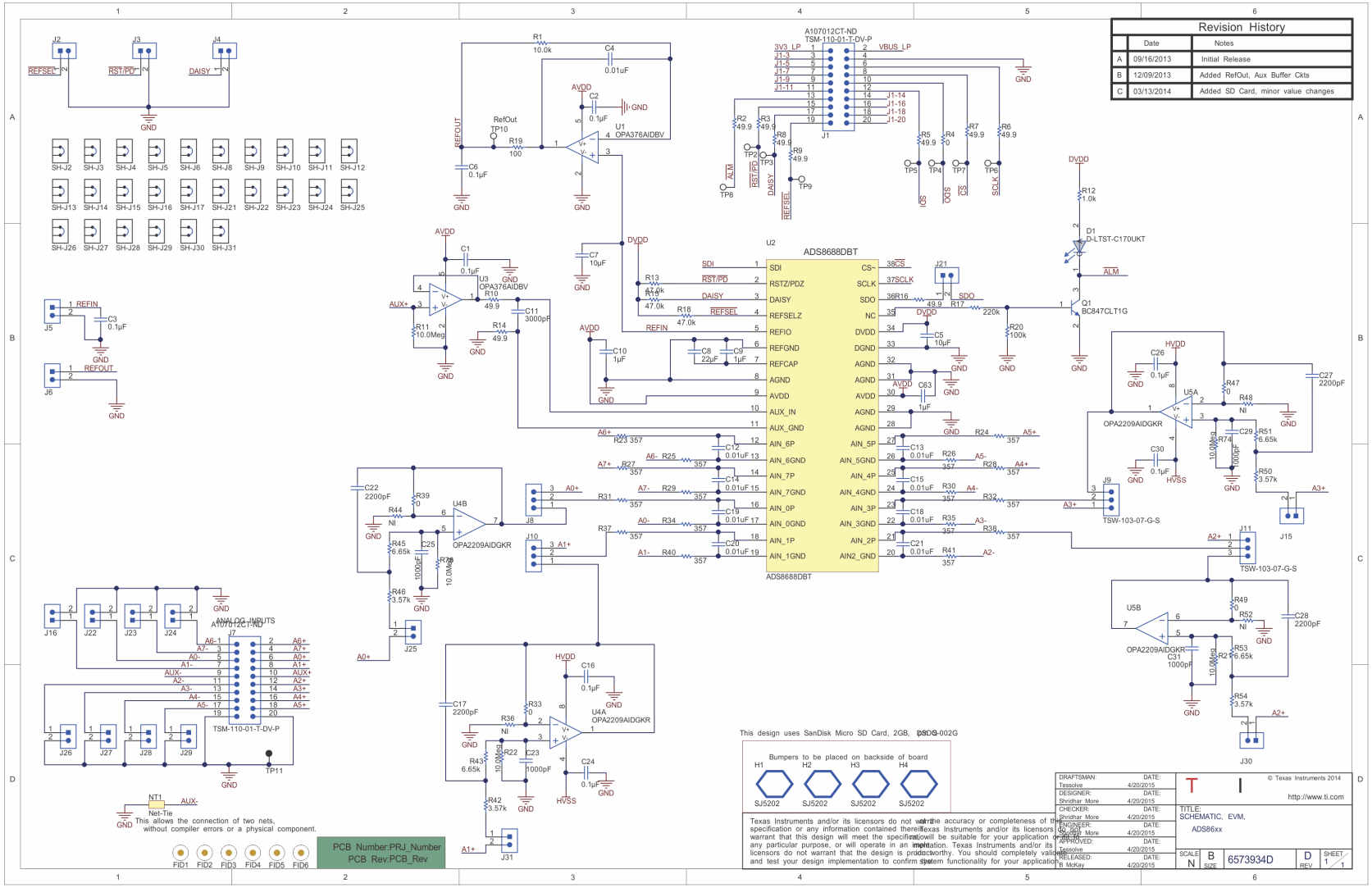


**Figure 36. ADS86xx Interface Schematic**



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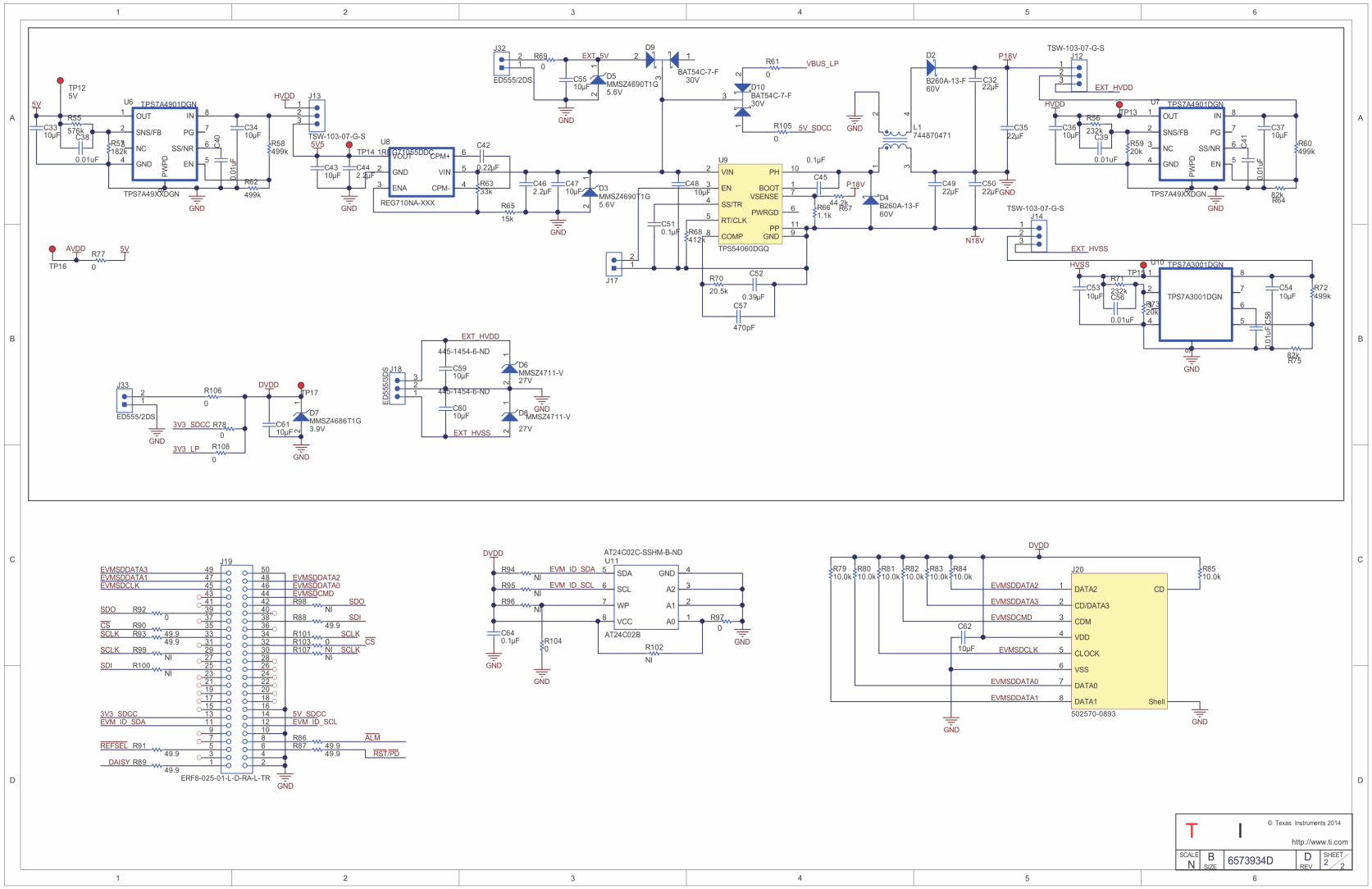


**Figure 37. Power and SDCC Interface Schematic**



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# Revision History

| Changes from Original (June 2015) to A Revision | Page |
|---|---|
| • Added title to *Related Documents* section | 4 |
| • Added reference to Section 8.3 in overview of Section 8 | 32 |
| • Added *Schematics* section | 39 |

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.


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1. *Delivery:* TI delivers TI evaluation boards, kits, or modules, including any accompanying demonstration software, components, or documentation (collectively, an “EVM” or “EVMs”) to the User (“User”) in accordance with the terms and conditions set forth herein. Acceptance of the EVM is expressly subject to the following terms and conditions.

   1.1 EVMs are intended solely for product or software developers for use in a research and development setting to facilitate feasibility evaluation, experimentation, or scientific analysis of TI semiconductors products. EVMs have no direct function and are not finished products. EVMs shall not be directly or indirectly assembled as a part or subassembly in any finished product. For clarification, any software or software tools provided with the EVM (“Software”) shall not be subject to the terms and conditions set forth herein but rather shall be subject to the applicable terms and conditions that accompany such Software

   1.2 EVMs are not intended for consumer or household use. EVMs may not be sold, sublicensed, leased, rented, loaned, assigned, or otherwise distributed for commercial purposes by Users, in whole or in part, or used in any finished product or production system.

2 *Limited Warranty and Related Remedies/Disclaimers*:

   2.1 These terms and conditions do not apply to Software. The warranty, if any, for Software is covered in the applicable Software License Agreement.

   2.2 TI warrants that the TI EVM will conform to TI's published specifications for ninety (90) days after the date TI delivers such EVM to User. Notwithstanding the foregoing, TI shall not be liable for any defects that are caused by neglect, misuse or mistreatment by an entity other than TI, including improper installation or testing, or for any EVMs that have been altered or modified in any way by an entity other than TI. Moreover, TI shall not be liable for any defects that result from User's design, specifications or instructions for such EVMs. Testing and other quality control techniques are used to the extent TI deems necessary or as mandated by government requirements. TI does not test all parameters of each EVM.

   2.3 If any EVM fails to conform to the warranty set forth above, TI's sole liability shall be at its option to repair or replace such EVM, or credit User's account for such EVM. TI's liability under this warranty shall be limited to EVMs that are returned during the warranty period to the address designated by TI and that are determined by TI not to conform to such warranty. If TI elects to repair or replace such EVM, TI shall have a reasonable time to repair such EVM or provide replacements. Repaired EVMs shall be warranted for the remainder of the original warranty period. Replaced EVMs shall be warranted for a new full ninety (90) day warranty period.

3 *Regulatory Notices:*

   3.1 *United States*

      3.1.1 *Notice applicable to EVMs not FCC-Approved:*

      This kit is designed to allow product developers to evaluate electronic components, circuitry, or software associated with the kit to determine whether to incorporate such items in a finished product and software developers to write software applications for use with the end product. This kit is not a finished product and when assembled may not be resold or otherwise marketed unless all required FCC equipment authorizations are first obtained. Operation is subject to the condition that this product not cause harmful interference to licensed radio stations and that this product accept harmful interference. Unless the assembled kit is designed to operate under part 15, part 18 or part 95 of this chapter, the operator of the kit must operate under the authority of an FCC license holder or must secure an experimental authorization under part 5 of this chapter.

      3.1.2 *For EVMs annotated as FCC – FEDERAL COMMUNICATIONS COMMISSION Part 15 Compliant:*

      **CAUTION**

      This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

      Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

      **FCC Interference Statement for Class A EVM devices**

      *NOTE: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.*

**FCC Interference Statement for Class B EVM devices**

*NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:*

- *Reorient or relocate the receiving antenna.*
- *Increase the separation between the equipment and receiver.*
- *Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.*
- *Consult the dealer or an experienced radio/TV technician for help.*

3.2 *Canada*

3.2.1 *For EVMs issued with an Industry Canada Certificate of Conformance to RSS-210*

**Concerning EVMs Including Radio Transmitters:**

This device complies with Industry Canada license-exempt RSS standard(s). Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

**Concernant les EVMs avec appareils radio:**

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

**Concerning EVMs Including Detachable Antennas:**

Under Industry Canada regulations, this radio transmitter may only operate using an antenna of a type and maximum (or lesser) gain approved for the transmitter by Industry Canada. To reduce potential radio interference to other users, the antenna type and its gain should be so chosen that the equivalent isotropically radiated power (e.i.r.p.) is not more than that necessary for successful communication. This radio transmitter has been approved by Industry Canada to operate with the antenna types listed in the user guide with the maximum permissible gain and required antenna impedance for each antenna type indicated. Antenna types not included in this list, having a gain greater than the maximum gain indicated for that type, are strictly prohibited for use with this device.

**Concernant les EVMs avec antennes détachables**

Conformément à la réglementation d'Industrie Canada, le présent émetteur radio peut fonctionner avec une antenne d'un type et d'un gain maximal (ou inférieur) approuvé pour l'émetteur par Industrie Canada. Dans le but de réduire les risques de brouillage radioélectrique à l'intention des autres utilisateurs, il faut choisir le type d'antenne et son gain de sorte que la puissance isotrope rayonnée équivalente (p.i.r.e.) ne dépasse pas l'intensité nécessaire à l'établissement d'une communication satisfaisante. Le présent émetteur radio a été approuvé par Industrie Canada pour fonctionner avec les types d'antenne énumérés dans le manuel d'usage et ayant un gain admissible maximal et l'impédance requise pour chaque type d'antenne. Les types d'antenne non inclus dans cette liste, ou dont le gain est supérieur au gain maximal indiqué, sont strictement interdits pour l'exploitation de l'émetteur

3.3 *Japan*

3.3.1 *Notice for EVMs delivered in Japan:* Please see http://www.tij.co.jp/lsds/ti_ja/general/eStore/notice_01.page 日本国内に輸入される評価用キット、ボードについては、次のところをご覧ください。
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3.3.2 *Notice for Users of EVMs Considered "Radio Frequency Products" in Japan:* EVMs entering Japan may not be certified by TI as conforming to Technical Regulations of Radio Law of Japan.

If User uses EVMs in Japan, not certified to Technical Regulations of Radio Law of Japan, User is required by Radio Law of Japan to follow the instructions below with respect to EVMs:

1. Use EVMs in a shielded room or any other test facility as defined in the notification #173 issued by Ministry of Internal Affairs and Communications on March 28, 2006, based on Sub-section 1.1 of Article 6 of the Ministry's Rule for Enforcement of Radio Law of Japan,
2. Use EVMs only after User obtains the license of Test Radio Station as provided in Radio Law of Japan with respect to EVMs, or
3. Use of EVMs only after User obtains the Technical Regulations Conformity Certification as provided in Radio Law of Japan with respect to EVMs. Also, do not transfer EVMs, unless User gives the same notice above to the transferee. Please note that if User does not follow the instructions above, User will be subject to penalties of Radio Law of Japan.

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4 *EVM Use Restrictions and Warnings:*

4.1 EVMS ARE NOT FOR USE IN FUNCTIONAL SAFETY AND/OR SAFETY CRITICAL EVALUATIONS, INCLUDING BUT NOT LIMITED TO EVALUATIONS OF LIFE SUPPORT APPLICATIONS.

4.2 User must read and apply the user guide and other available documentation provided by TI regarding the EVM prior to handling or using the EVM, including without limitation any warning or restriction notices. The notices contain important safety information related to, for example, temperatures and voltages.

4.3 *Safety-Related Warnings and Restrictions:*

4.3.1 User shall operate the EVM within TI's recommended specifications and environmental considerations stated in the user guide, other available documentation provided by TI, and any other applicable requirements and employ reasonable and customary safeguards. Exceeding the specified performance ratings and specifications (including but not limited to input and output voltage, current, power, and environmental ranges) for the EVM may cause personal injury or death, or property damage. If there are questions concerning performance ratings and specifications, User should contact a TI field representative prior to connecting interface electronics including input power and intended loads. Any loads applied outside of the specified output range may also result in unintended and/or inaccurate operation and/or possible permanent damage to the EVM and/or interface electronics. Please consult the EVM user guide prior to connecting any load to the EVM output. If there is uncertainty as to the load specification, please contact a TI field representative. During normal operation, even with the inputs and outputs kept within the specified allowable ranges, some circuit components may have elevated case temperatures. These components include but are not limited to linear regulators, switching transistors, pass transistors, current sense resistors, and heat sinks, which can be identified using the information in the associated documentation. When working with the EVM, please be aware that the EVM may become very warm.

4.3.2 EVMs are intended solely for use by technically qualified, professional electronics experts who are familiar with the dangers and application risks associated with handling electrical mechanical components, systems, and subsystems. User assumes all responsibility and liability for proper and safe handling and use of the EVM by User or its employees, affiliates, contractors or designees. User assumes all responsibility and liability to ensure that any interfaces (electronic and/or mechanical) between the EVM and any human body are designed with suitable isolation and means to safely limit accessible leakage currents to minimize the risk of electrical shock hazard. User assumes all responsibility and liability for any improper or unsafe handling or use of the EVM by User or its employees, affiliates, contractors or designees.

4.4 User assumes all responsibility and liability to determine whether the EVM is subject to any applicable international, federal, state, or local laws and regulations related to User's handling and use of the EVM and, if applicable, User assumes all responsibility and liability for compliance in all respects with such laws and regulations. User assumes all responsibility and liability for proper disposal and recycling of the EVM consistent with all applicable international, federal, state, and local requirements.

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